UNIVERSITY OF AUCKLAND
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SCIENCE

COMPSCI 714
Lecture 9: Transfer Learning III

Pretrained Foundational Models

Instructor: Qian Liu

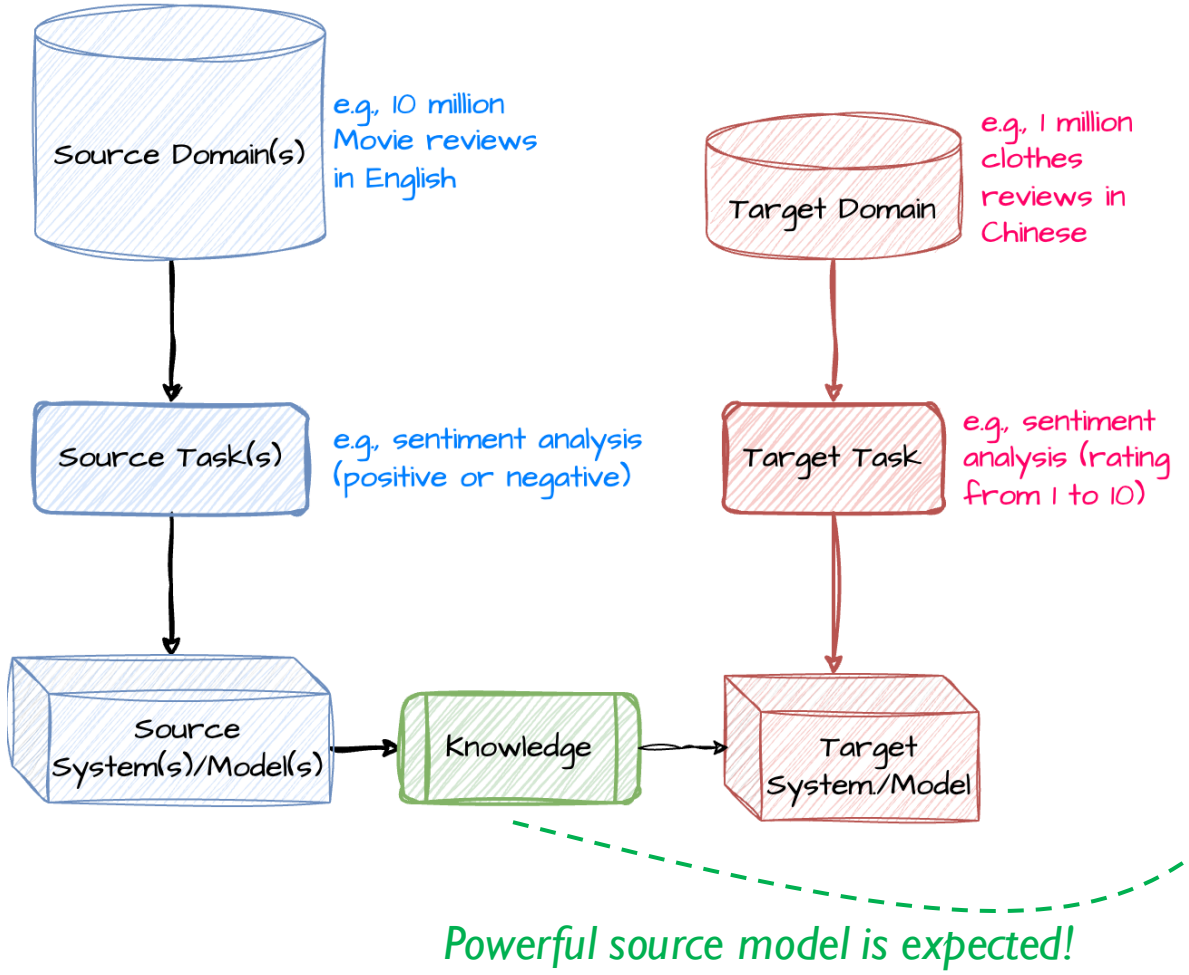
Outline

1. Overview of Pre-training and Fine-tuning Paradigm
2. Some important concepts
   1. Feature
   2. Representation learning
   3. Encoder and Decoder
3. Pretrained Foundation Models in NLP
   a. Encoders (BERT)
   b. Encoder-Decoders (T5)
   c. Decoders (GPT-1)
4. Pretrained Foundation Models in CV (This Friday)

Assignment-3 is released.

Due on end of Week 9

Transfer Learning



For source domain, how to train a powerful model?

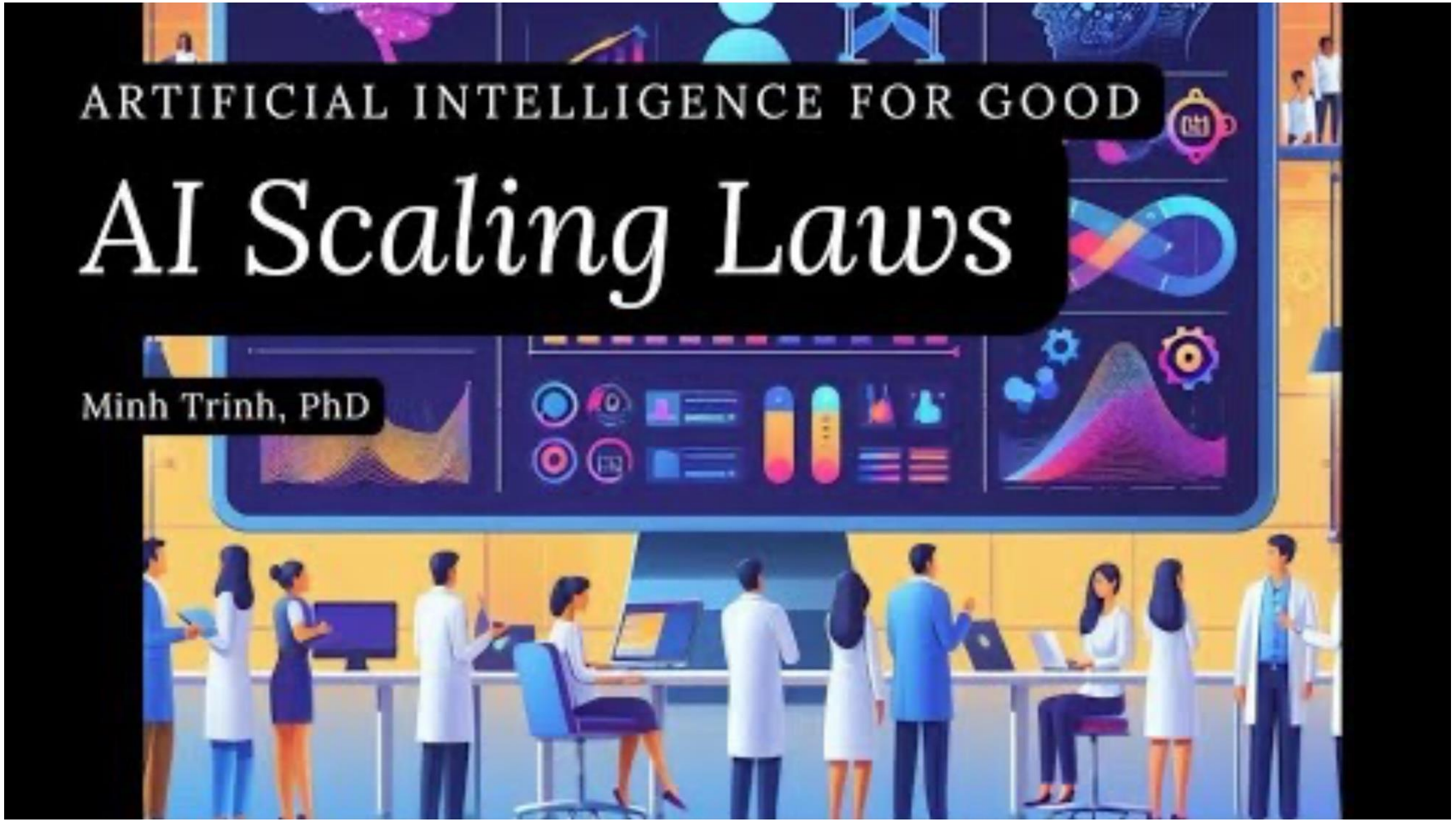
AI scaling law refers to an empirical principle that describes how the performance of AI models—especially large neural networks like transformers—**improves in a predictable way** as you scale up key resources:

🔑 **Key Resources in AI Scaling Laws:**

1. **Model size** — number of parameters
2. **Dataset size** — how much data the model is trained on
3. **Compute budget** — total computation used for training (FLOPs)

Bigger models trained on more data with more compute resources will perform better

AI Scaling Law!



Scaling Laws for Neural Language Models

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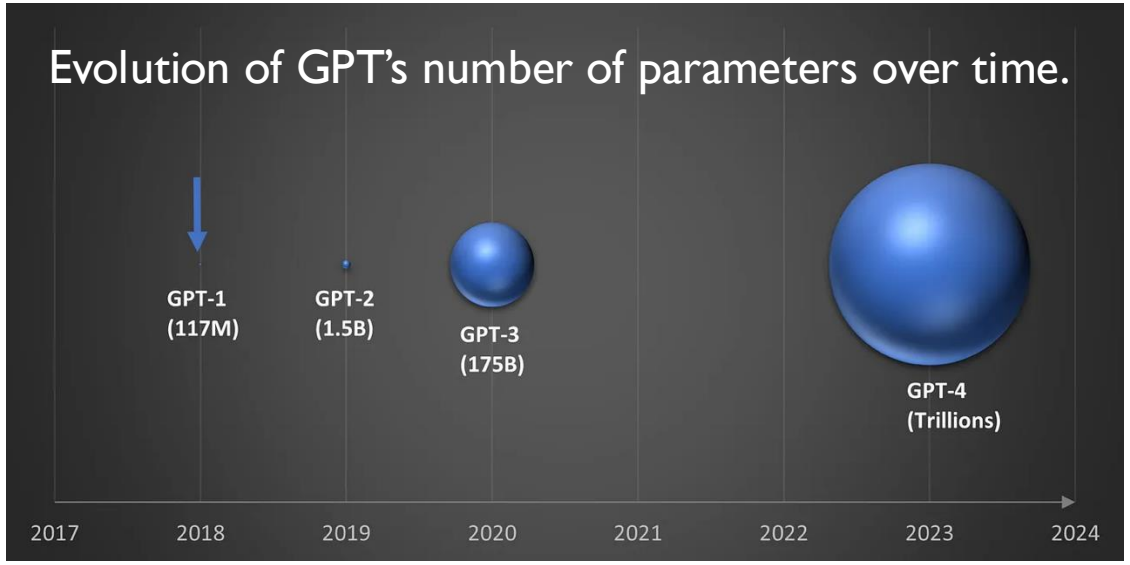
Abstract

We study empirical scaling laws for language model performance on the cross-entropy loss. The loss scales as a power-law with model size, dataset size, and the amount of compute used for training, with some trends spanning more than seven orders of magnitude. Other architectural details such as network width or depth have minimal effects within a wide range. Simple equations govern the dependence of overfitting on model/dataset size and the dependence of training speed on model size. These relationships allow us to determine the optimal allocation of a fixed compute budget. Larger models are significantly more sample-efficient, such that optimally compute-efficient training involves training very large models on a relatively modest amount of data and stopping significantly before convergence.

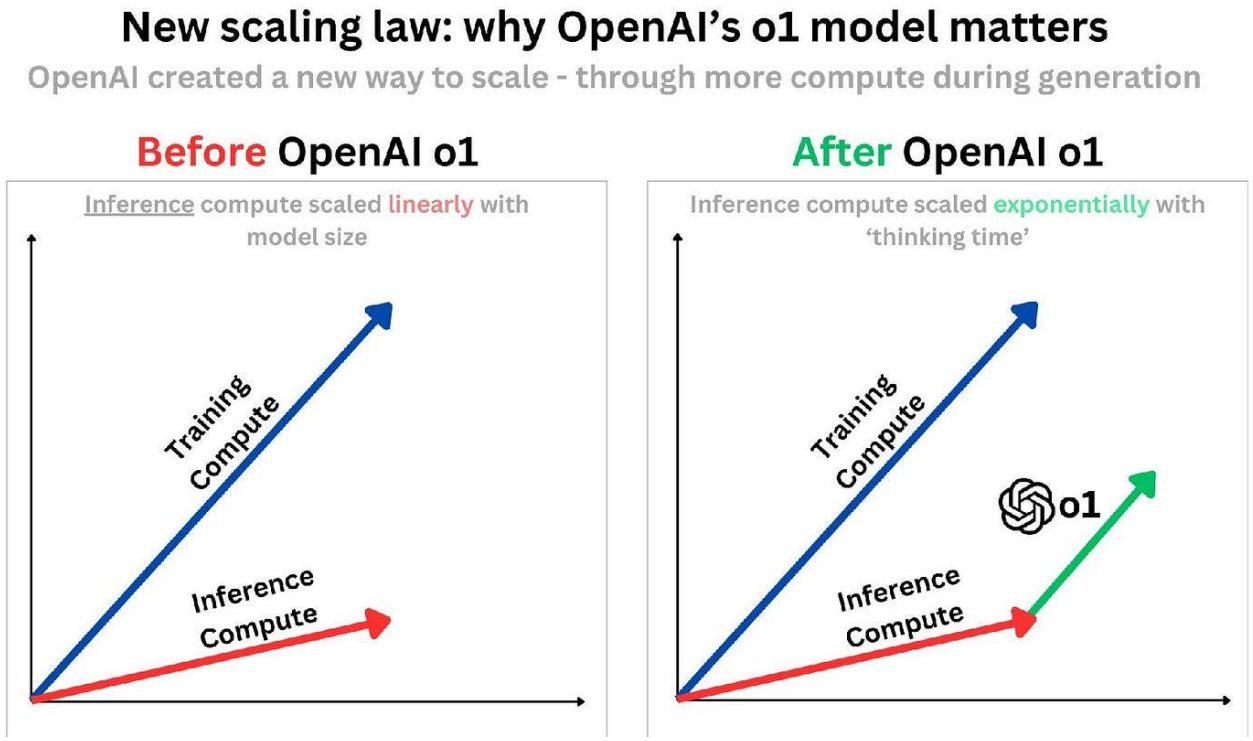
https://arxiv.org/abs/2001.08361

bigger is better!

AI Scaling Law!

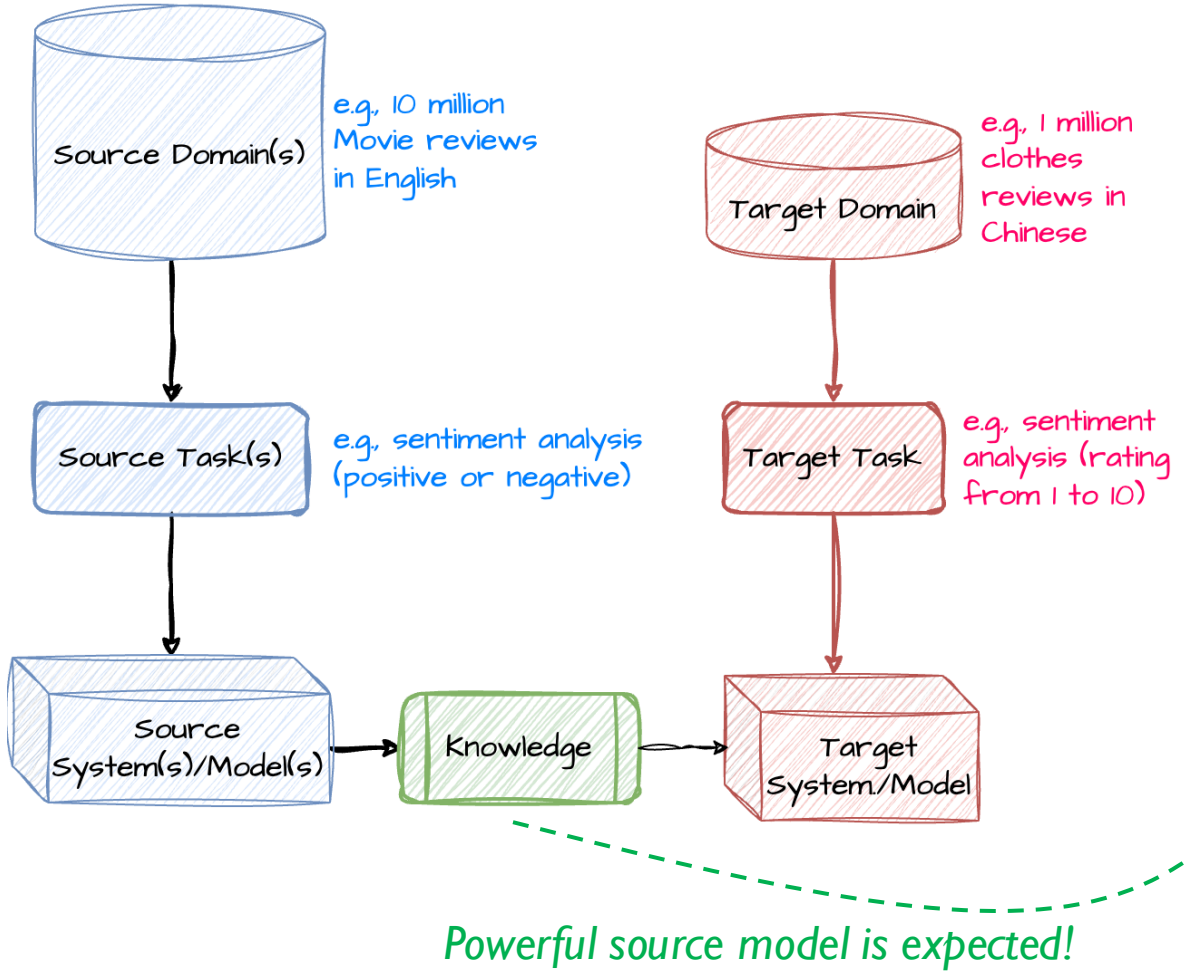


Model	Parameters	Dataset Size (tokens)
GPT-2 small	124M	0.5B
GPT-2 medium	345M	1.5B
GPT-2 large	762M	3B
GPT-2 XL	1.5B	5B
GPT-3	175B	300B



https://medium.com/@aiml_58187/beyond-bigger-models-the-evolution-of-language-model-scaling-laws-d4bc974d3876

Transfer Learning



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If we aim to train a large model on large-scale data, relying on labeled data alone is virtually impossible due to the high cost and limited availability of annotations.

Transfer Learning

		Source Data	
		labelled	unlabeled
Target Data	labelled	Model Fine-tuning Multitask Learning	**Self-supervised Learning**
	unlabeled	Domain-adversarial training Zero/Few-shot learning	

If we expect a big model trained on large-scale data, it is almost impossible to use labeled data

↓

Unlabeled data is much more **plentiful and easier to obtain** than labeled data.

↓

Modern transfer learning heavily relies on **self-supervised learning** — where **pseudo-tasks** (like masked language modeling, next sentence prediction, contrastive learning) are used to train models on unlabeled data. These tasks help models build useful internal representations without requiring manual labels.

Self-supervised Learning

- **Self-supervised learning** (**SSL**) is a paradigm in machine learning where a model is trained on a task using the data itself to generate supervisory signals, rather than relying on externally-provided labels.

cat

dog

tiger

Images with labels
(manually annotated,
harder to obtain)

Images without labels
(plentiful and easier to obtain)

Step-1: It is easy to collect millions of colourful images

Step-2: Converting a colorful image to a grey image

Self-supervised Learning

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cat

dog

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Images with labels
(manually annotated,
harder to obtain)

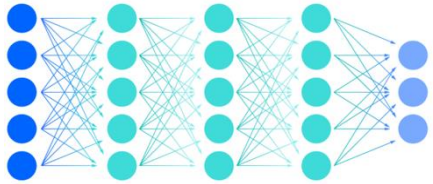
Original image

Original image works as the label for this task

- **Input:** Grayscale version of a colorful image (color is removed).
- **Objective:** Predict the original color version from the grayscale.

No manually labels are required!

input



model

Output
(predicted color)

Self-supervised Learning

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Piece of subtle art. Maybe a masterpiece. Doubtlessly a special story about the ambiguity of existence. Tale in Kafka style about impossibility of victory or surviving in a perpetual strange world. The life is, in this film, only exercise of adaptation. Lesson about limits and original sin, about the frailty of innocence and error of his ways. Leopold Kessle is another Joseph K. Images of Trial and same ambiguous woman. And Europa is symbol of basic crisis who has many aspects like chimeric wars or unavailing search of truth/essence/golden age. Methaphor or parable, the movie is history of disappointed's evolution. War, peace, business or lie are only details of gelatin-time. Hypocrisy is a mask. Love- a convention. The sacrifice- only method to hope understanding a painful reality.	1 pos
If only to avoid making this type of film in the future. This film is interesting as an experiment but tells no cogent story. One might feel virtuous for sitting thru it because it touches on so many IMPORTANT issues but it does so without any discernable motive. The viewer comes away with no new perspectives (unless one comes up with one while one's mind wanders, as it will invariably do during this pointless film). One might better spend one's time staring out a window at a tree growing. 	0 neg

Labels

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No label. Easy to obtain!

Randomly mask some words

If only to avoid [redacted] this type of film in the future. This film is interesting as an experiment but tells no cogent story.

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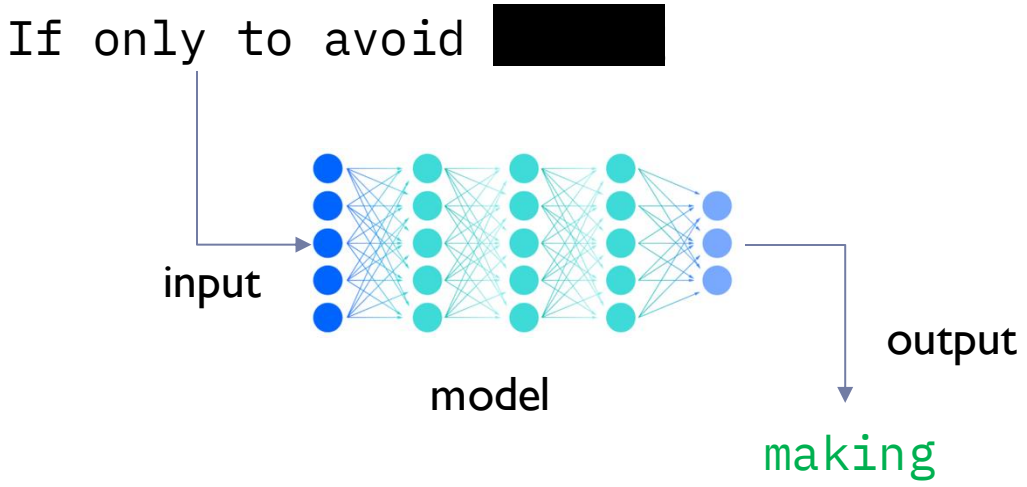
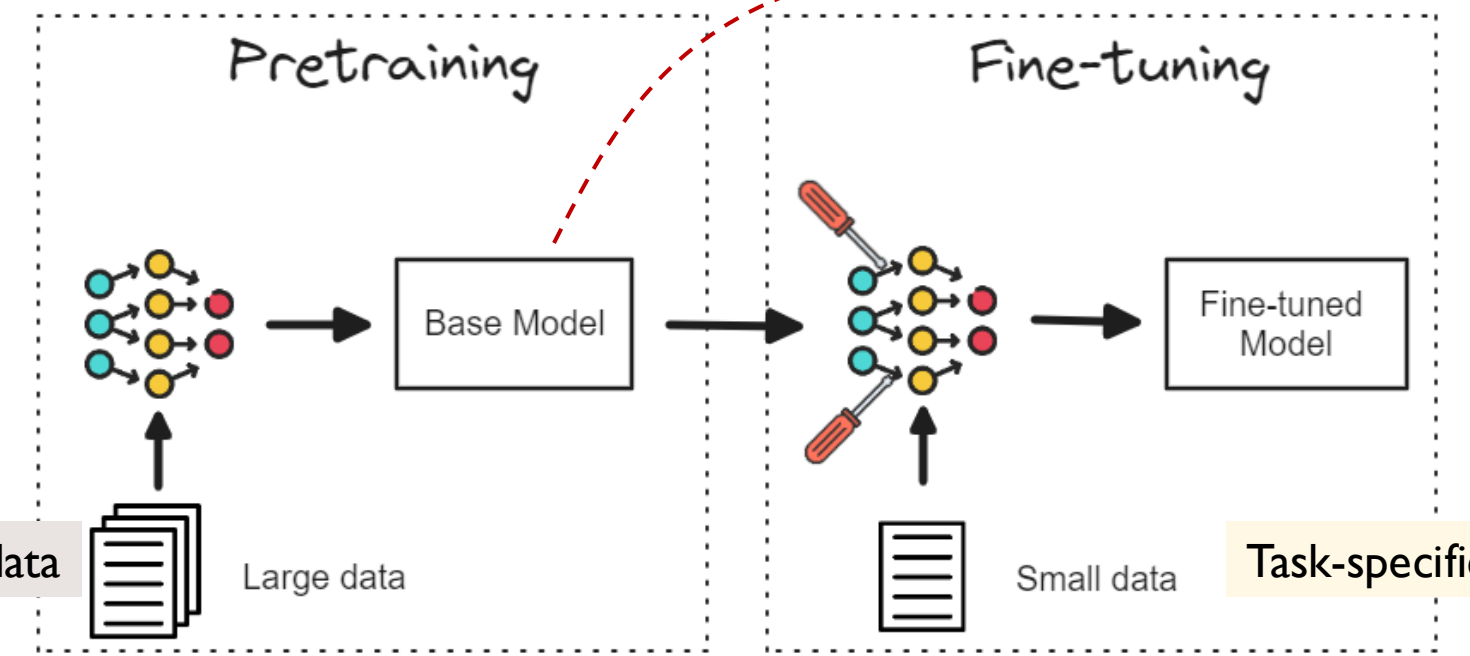


*The model's task is to **predict the original tokens** that were masked*
*The **supervision signal comes from the data itself***

Pre-training and Fine-tuning Paradigm

The pre-trained models are denoted as base models/ foundational models

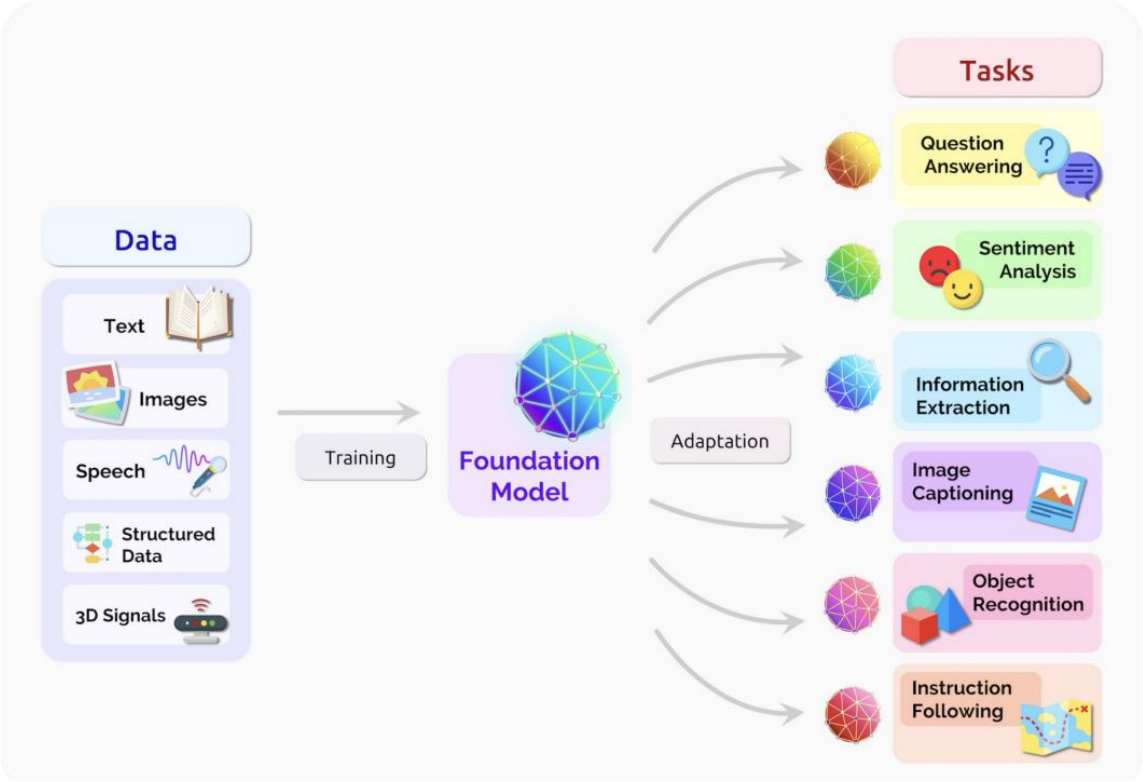


Pre-training: A model learns general-purpose knowledge from a large amount of unlabeled data in a self-supervised way.

Fine-tuning: Fine-tune the pre-trained model and adjust it for a specific task using small-scale labeled data.

Pretrained Foundation Models (PFMs)

- Pretrained Foundation Models (PFMs) are regarded as the foundation for various downstream tasks. A PFM is trained on large-scale data which provides a reasonable parameter initialization for a wide range of downstream applications.



Key ideas in pretraining

- Make sure your model can process large-scale, diverse datasets
- Don't use labeled data (otherwise you can't scale!)
- Compute-aware scaling

Analogy

- **Pre-training** is like attending school and learning general knowledge.
- **Fine-tuning** is like job training for a specific profession.

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3. Pretrained Foundation Models in NLP
   a. Encoders (BERT)
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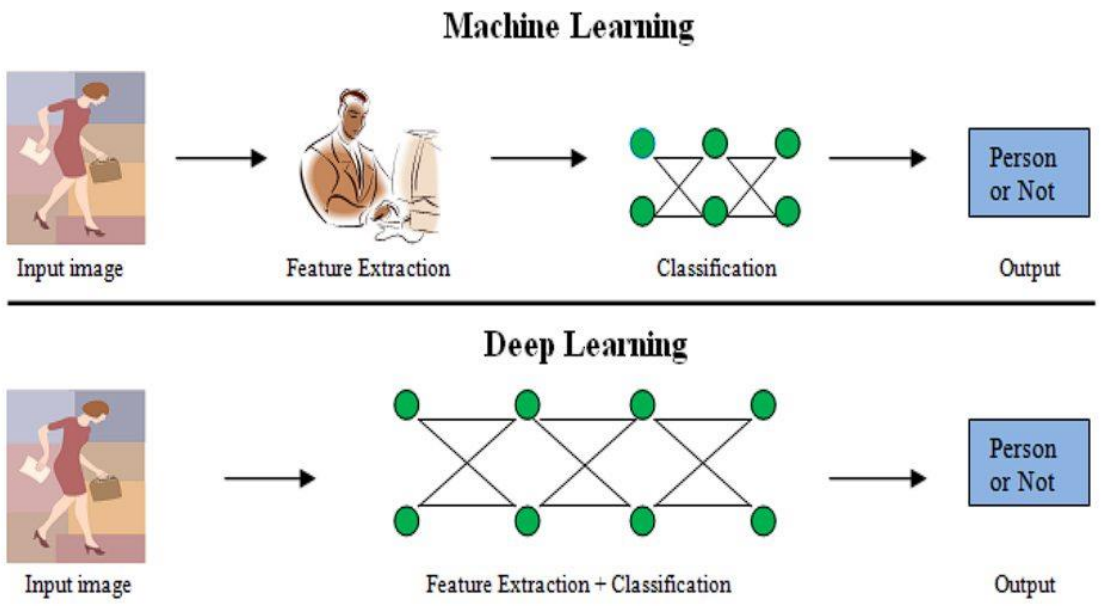
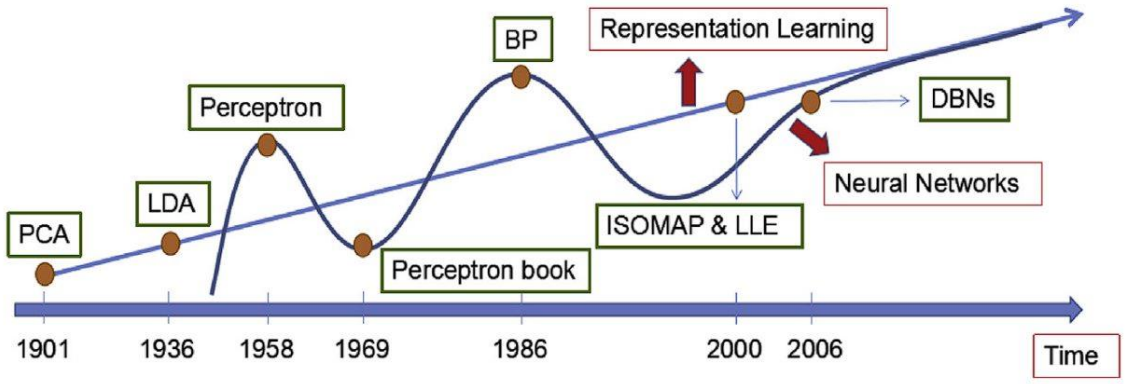
Some important concepts

- **Feature**
  - AI models, especially machine learning models, **cannot** directly interpret raw data such as text, images, or audio in their original form.
  - Instead, they require input data to be converted into numerical representations—called features—that capture the relevant patterns or properties of the data.





Representation Learning Timeline

https://viso.ai/deep-learning/representation-learning/

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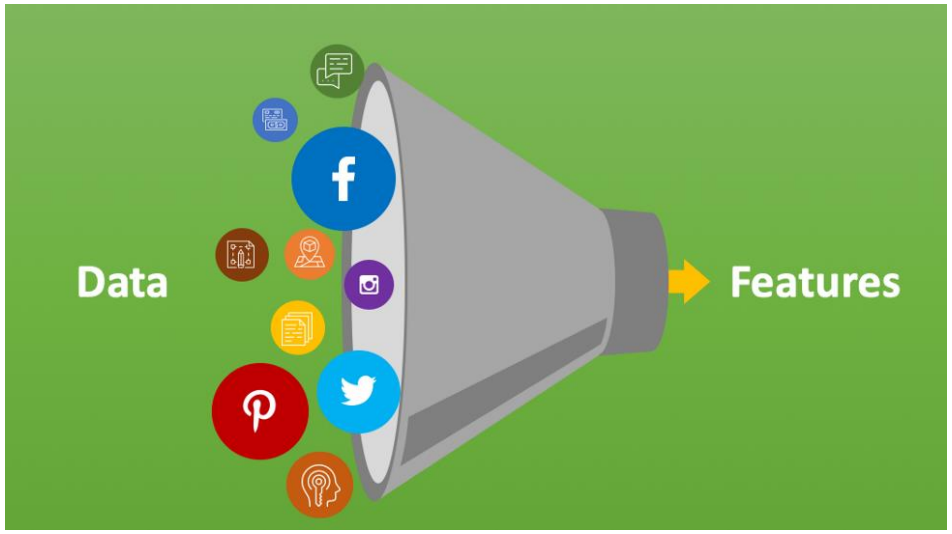

For text data: "We convert words into embeddings—vectors of numbers—that reflect semantic meaning and context."
For image data: "We represent an image as a matrix of pixel values or extract higher-level features like edges and textures using convolutional layers."

Widely denoted as Feature Learning or Representation Learning

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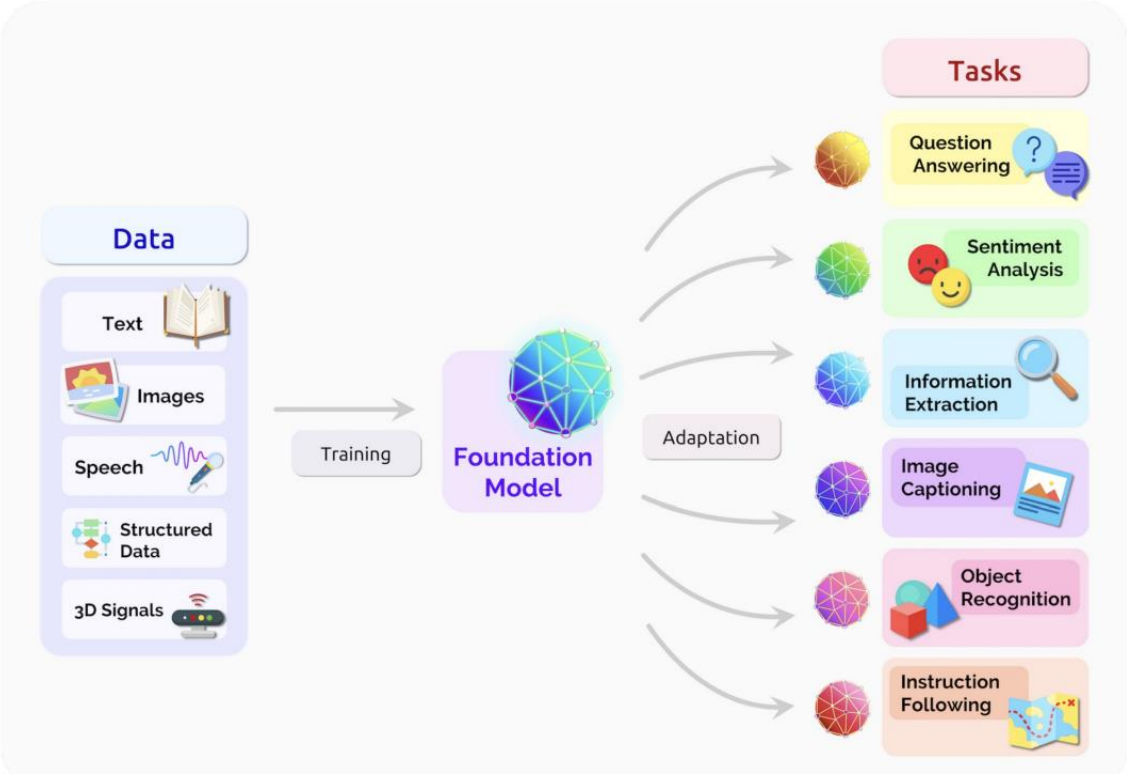
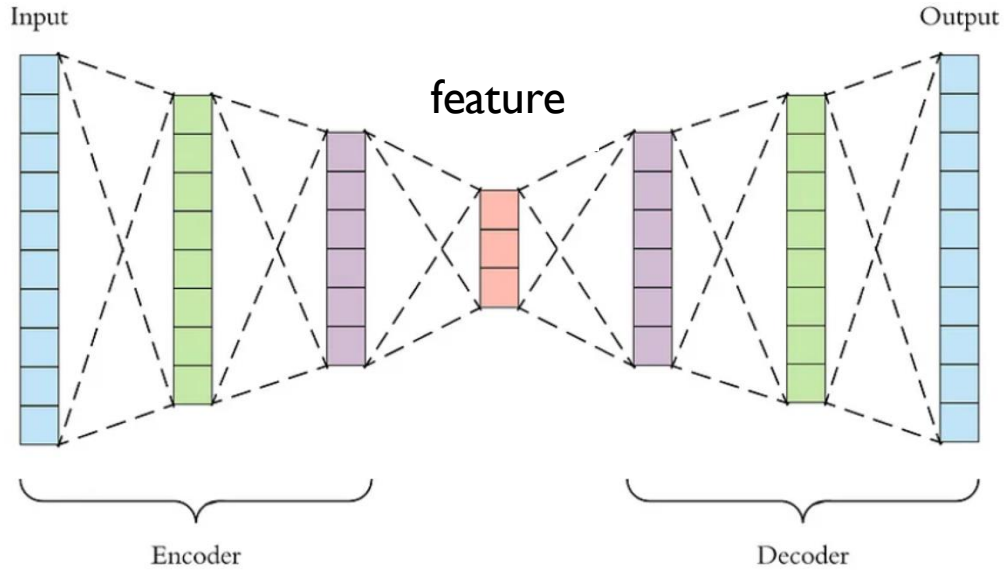


Pretrained Foundational Models are powerful feature extractor

1. High performance to represent input data as features
2. Avoid training from scratch

Some important concepts

- Feature
- Representation Learning
- Embedding
- **Encoder and Decoder**




```
Input → [Encoder] → Latent Representation → [Decoder] → Output
```

An **encoder** is a module that takes **input data** (like a sentence or an image) and transforms it into a **hidden representation** (often a vector or a set of vectors). The idea is to capture the **essential features or meaning** of the input.

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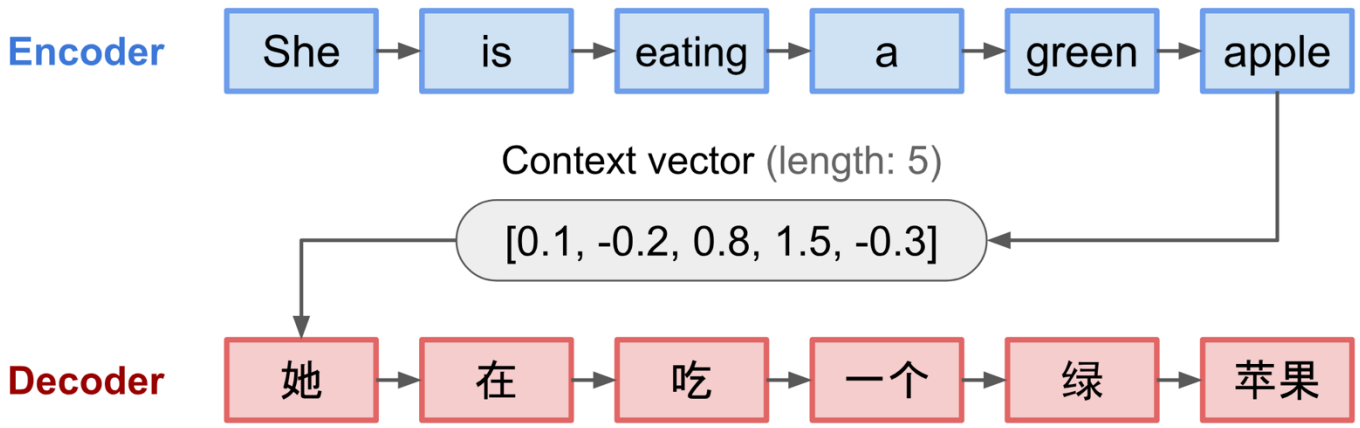

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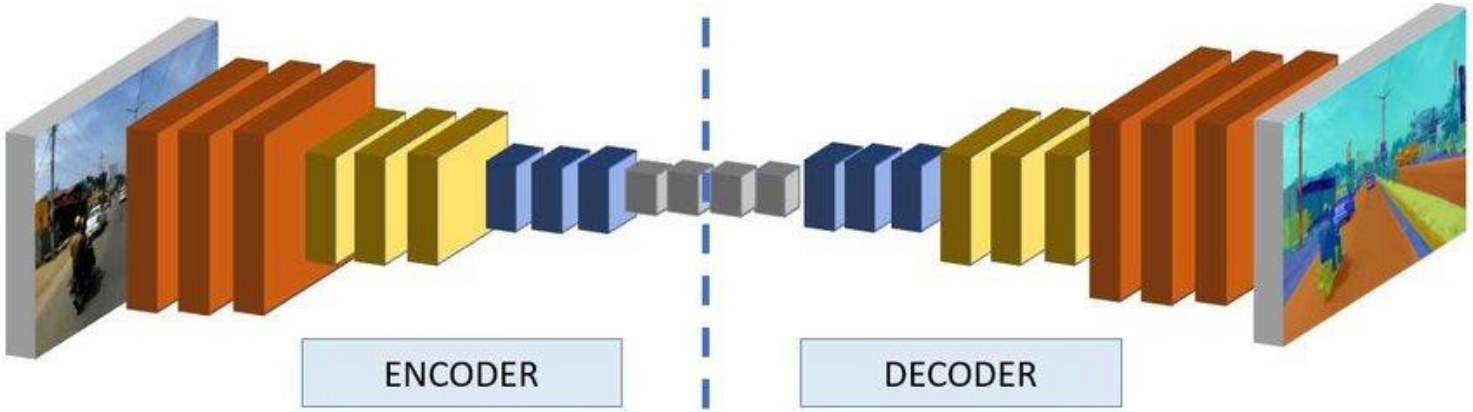

Input image | Latent representation | Output image (segmentation)

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Language Modeling

Most popular self-supervised learning in NLP

- A language model is a probabilistic model of a natural language.

- In NLP, the language modeling task is to predict the next word given a sequence of words. It is to model the probability distribution over words given their past context $p_\theta(w_t|w_1, w_2, w_3 \dots w_{t-1})$

- **Pretraining through language modeling:**
  Train a model (e.g., neural networks) to perform language modeling on a large amount of text

S = Where are we going

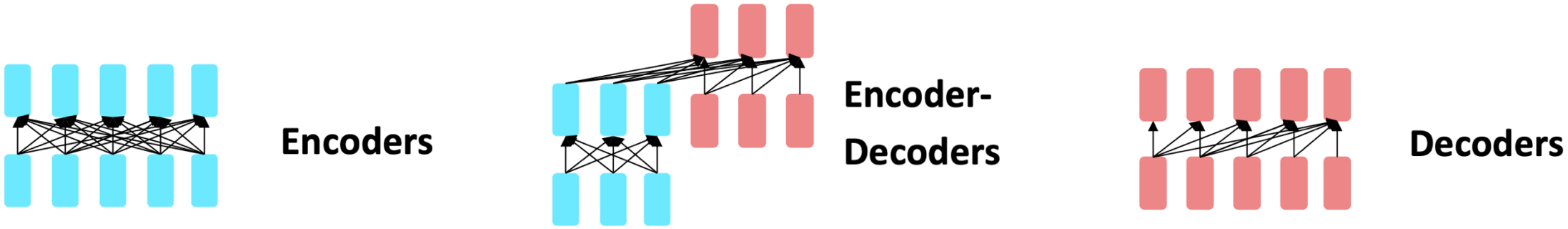
Previous words (Context)

Word being predicted

P(S) = P(Where) x P(are | Where) x P(we | Where are) x P(going | Where are we)

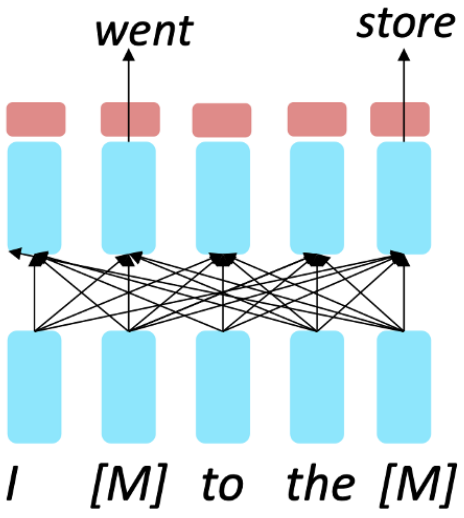
Three architectures of pretraining in NLP

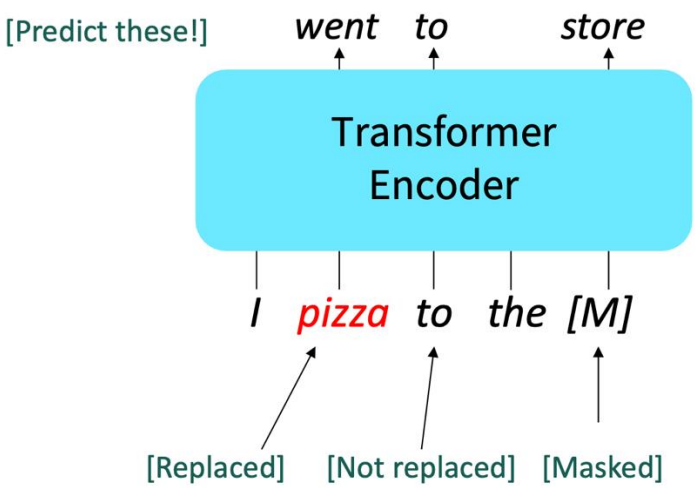
- The neural architecture influences the type of pretraining, and natural use cases.



Pretraining Encoders--BERT

- BERT: Bidirectional Encoder Representations from Transformers
-
- Training objectives:
  - Masked Language Modeling is to predict a random 15% of tokens
    - Replace input word with [MASK], 80% of the time
    - Replace input word with a random token, 10% of the time
    - Leave input word unchanged but still predict it, 10% of the time
  - Next Sentence Prediction
    - To predict whether one sentence follows the other or is randomly sampled.





https://arxiv.org/pdf/1810.04805

Pretraining Encoders

- Details of BERT
  - Two models were released:
    - BERT-base: 12 layers, 768-dim hidden states, 12 attention heads, 110 million params.
    - BERT-large: 24 layers, 1024-dim hidden states, 16 attention heads, 340 million params.
  - Trained on:
    - BooksCorpus (800 million words)
    - English Wikipedia
  - Pretraining is expensive and impractical on a single GPU.
    - BERT was pretrained with 64 TPU chips for a total of 4 days
    - (TPUs are special tensor operation acceleration hardware)
  - Finetuning is practical and common on a single GPU
    - "Pretrain once, finetune many times."

- **QQP:** Quora Question Pairs (detect paraphrase questions)
- **QNLI**: natural language inference over question answering data
- **SST-2**: sentiment analysis
- **CoLA**: corpus of linguistic acceptability (detect whether sentences are grammatical.)
- **STS-B**: semantic textual similarity
- **MRPC**: microsoft paraphrase corpus
- **RTE**: a small natural language inference corpus

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	**Average** -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
$BERT_{BASE}$	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
$BERT_{LARGE}$	**86.7/85.9**	**72.1**	**92.7**	**94.9**	**60.5**	**86.5**	**89.3**	**70.1**	**82.1**

finetuning BERT led to new state-of- the-art results on a broad range of tasks.

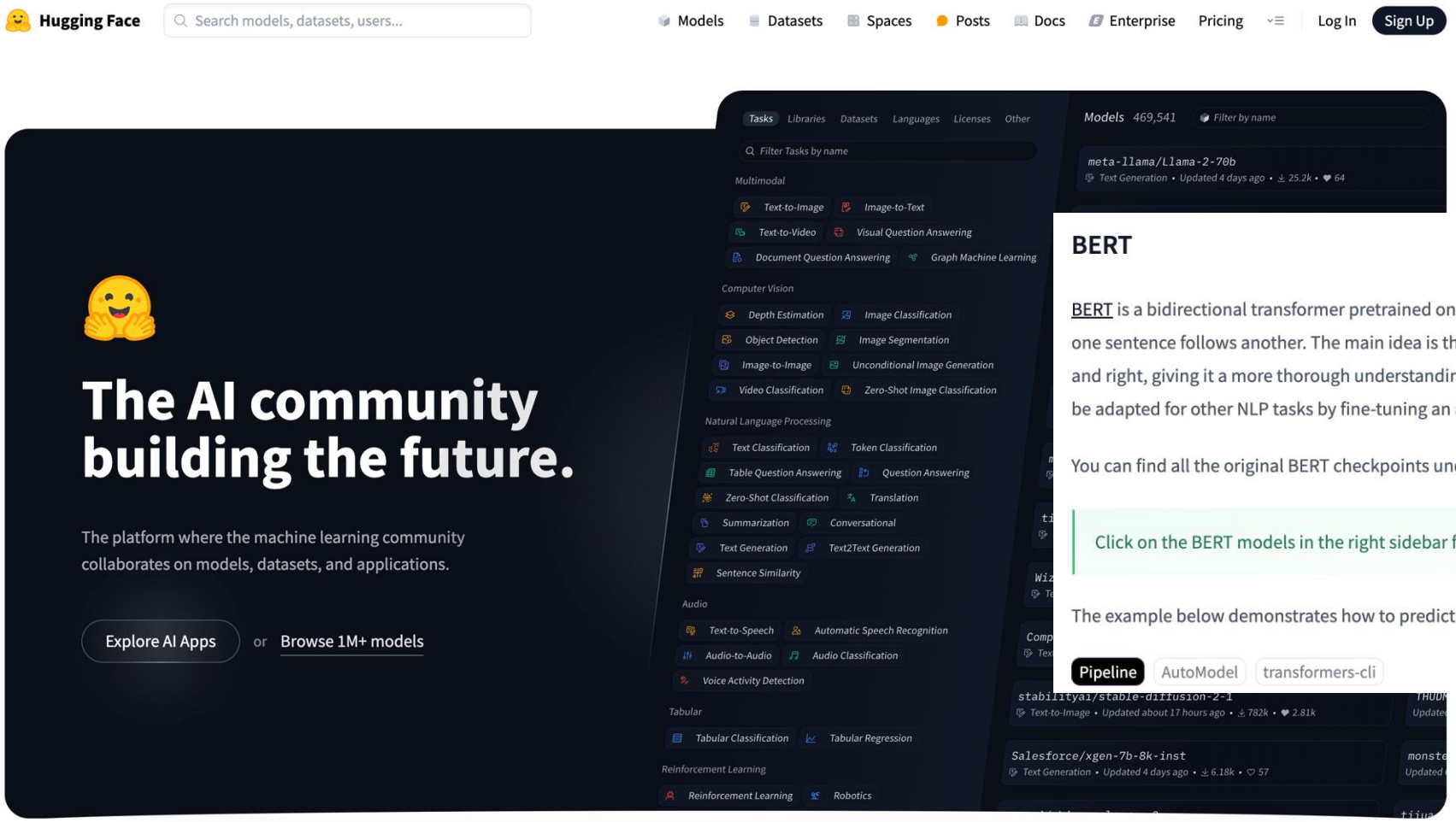
https://arxiv.org/pdf/1810.04805

Pretraining Encoders

- Extension of BERT
  - RoBERTa: mainly just train BERT for longer and remove next sentence prediction!
  - A takeaway from the RoBERTa paper: more compute, more data can improve pretraining even when not changing the underlying Transformer encoder.

Comparison	BERT October 11, 2018	RoBERTa July 26, 2019	DistilBERT October 2, 2019	ALBERT September 26, 2019
Parameters	**Base:** 110M **Large:** 340M	**Base:** 125 **Large:** 355	**Base:** 66	**Base:** 12M **Large:** 18M
Layers / Hidden Dimensions / Self-Attention Heads	**Base:** 12 / 768 / 12 **Large:** 24 / 1024 / 16	**Base:** 12 / 768 / 12 **Large:** 24 / 1024 / 16	**Base:** 6 / 768 / 12	**Base:** 12 / 768 / 12 **Large:** 24 / 1024 / 16
Training Time	**Base:** 8 x V100 x 12d **Large:** 280 x V100 x 1d	1024 x V100 x 1 day (4-5x more than BERT)	**Base:** 8 x V100 x 3.5d (4 times less than BERT)	[not given] **Large:** 1.7x faster
Performance	Outperforming SOTA in Oct 2018	88.5 on GLUE	97% of BERT-base's performance on GLUE	89.4 on GLUE
Pre-Training Data	BooksCorpus + English Wikipedia = 16 GB	BERT + CCNews + OpenWebText + Stories = 160 GB	BooksCorpus + English Wikipedia = 16 GB	BooksCorpus + English Wikipedia = 16 GB
Method	Bidirectional Transformer, MLM & NSP	BERT without NSP, Using Dynamic Masking	BERT Distillation	BERT with reduced parameters & SOP (not NSP)

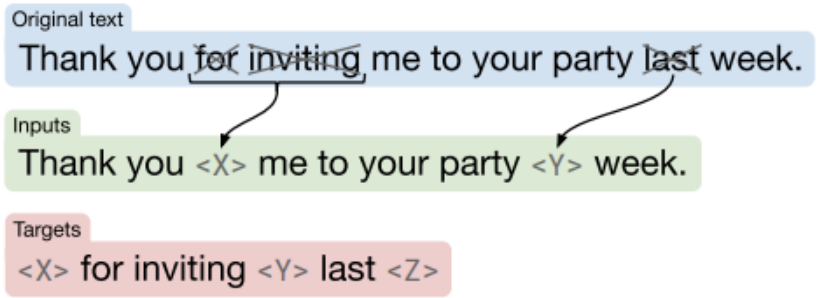
How to use BERT?



https://huggingface.co/docs/transformers/en/model_doc/bert

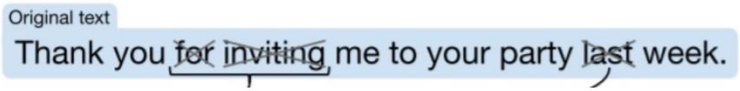
https://huggingface.co/

Pretraining Encoder-Decoders – T5

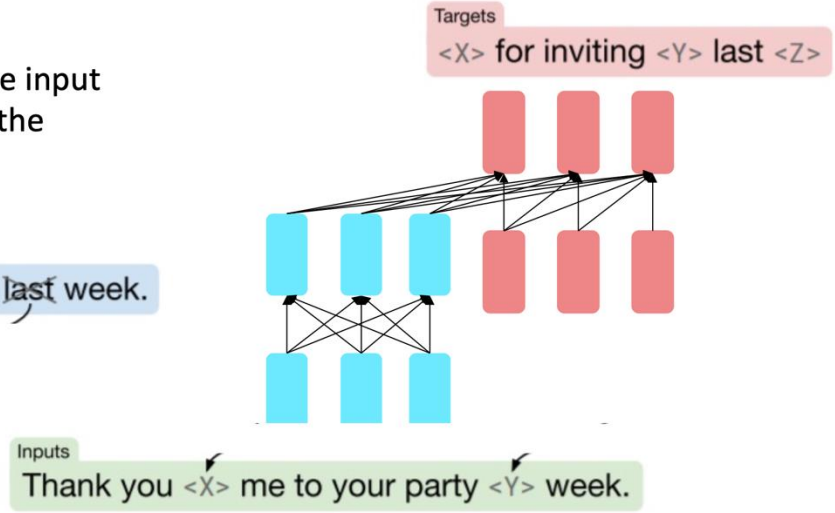


- T5 (Text-To-Text Transfer Transformer)
- Pretraining objective: Span Corruption
  - Input: A corrupted version of a text sequence, where **random contiguous spans** of tokens are **masked and replaced** with a single special token
  - Output: The model is trained to generate the **missing spans** in order

Replace different-length spans from the input with unique placeholders; decode out the spans that were removed!



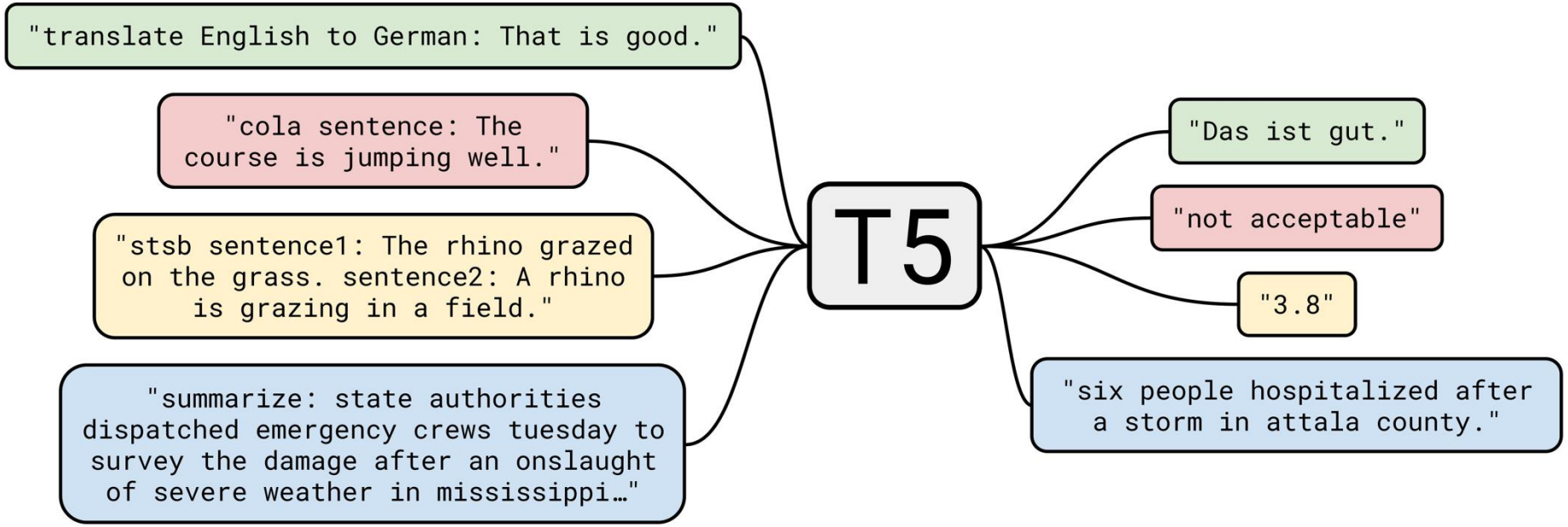
This is implemented in text preprocessing: it's still an objective that looks like **language modeling** at the decoder side.



- Encourages **long-range understanding** and richer representations.
- More suitable for **sequence-to-sequence** tasks (e.g., summarization, translation).
- Matches the **unified text-to-text format**: all tasks (translation, classification, question answering) are posed as **text-in → text-out**.

https://arxiv.org/pdf/1910.10683.pdf

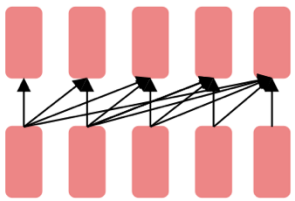
Pretraining Encoder-Decoders



https://arxiv.org/pdf/1910.10683.pdf

Pretraining Decoders– GPT

- **GPT (Generative Pre-trained Transformer)**
  - It is designed to generate human-like text by predicting the next word (token) in a sequence.
  - It is decoder-only architecture
  - Training objective: Causal Language Modeling (next-token prediction)



Decoders

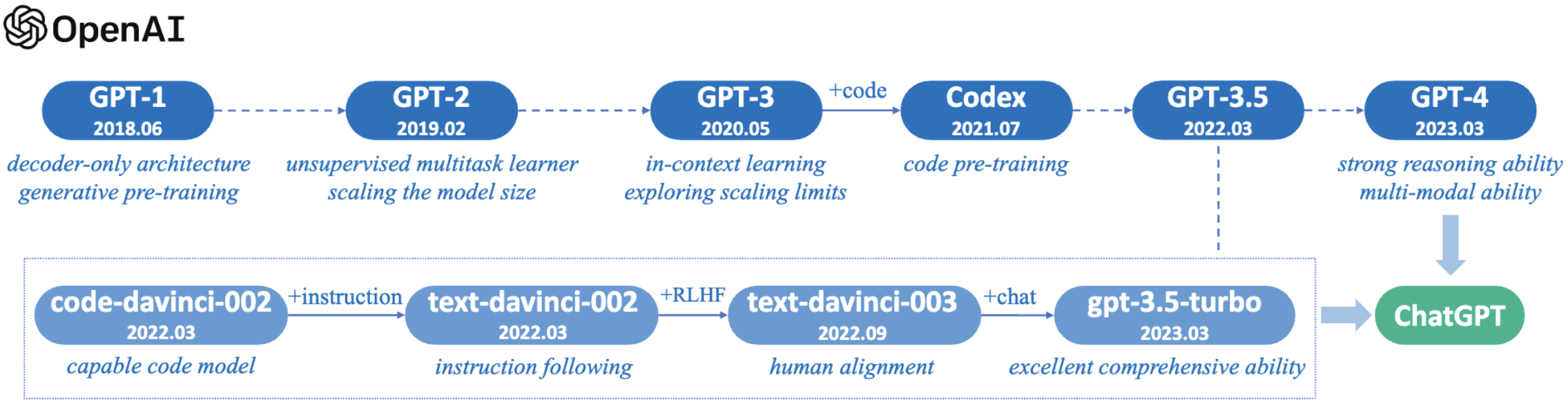
2018's GPT was a big success in pretraining a decoder!

- Transformer decoder with 12 layers, 117M parameters.
- 768-dimensional hidden states, 3072-dimensional feed-forward hidden layers.
- Byte-pair encoding with 40,000 merges
- Trained on BooksCorpus: over 7000 unique books.
  - Contains long spans of contiguous text, for learning long-distance dependencies.
- The acronym "GPT" never showed up in the original paper; it could stand for "Generative PreTraining" or "Generative Pretrained Transformer"

https://www.cs.ubc.ca/~amuham01/LING530/papers/radford2018improving.pdf

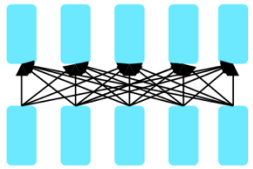
Pretrained Decoders

- A brief illustration for the technical evolution of GPT-series models. We plot this figure mainly based on the papers, blog articles and official APIs from OpenAI. Here, solid lines denote that there exists an explicit evidence (e.g., the official statement that a new model is developed based on a base model) on the evolution path between two models, while dashed lines denote a relatively weaker evolution relation.



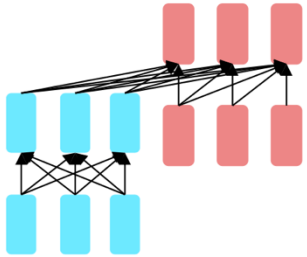
Pretrained Models in NLP

- Three types of architectures for pretraining: the neural architecture influences the type of pretraining, and natural use cases.



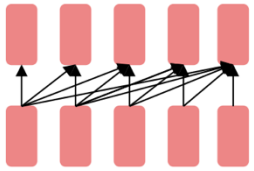
Encoders

- Gets bidirectional context
- Good if only doing analysis of text (better than decoders)
- Examples: BERT…



Encoder-Decoders

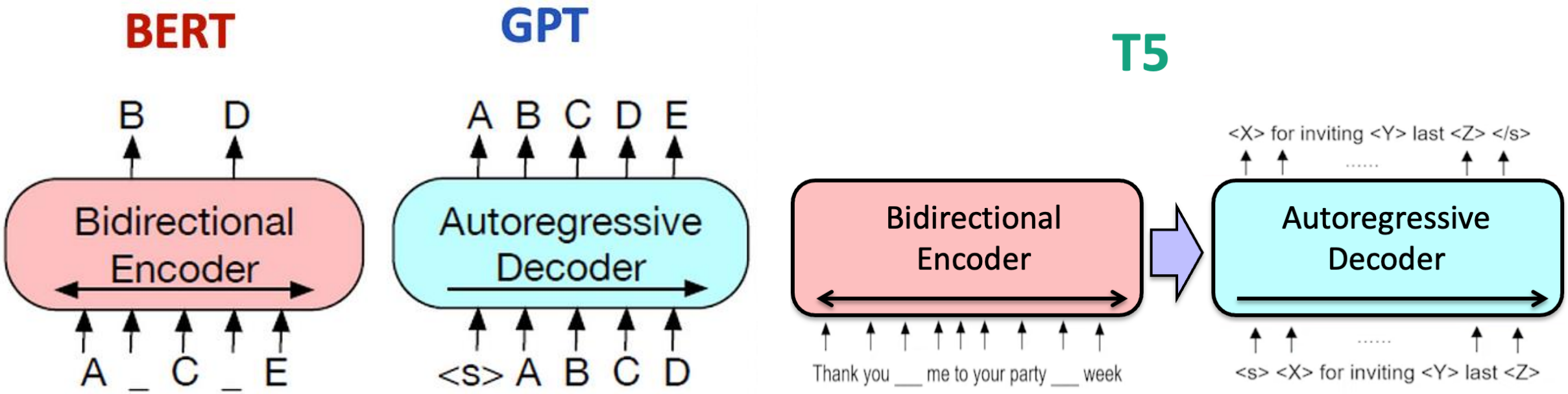
- Good parts of encoders and decoders
- Some evidence they are better for natural language understanding tasks
- Examples: T5…



Decoders

- Language models, scale well.
- Best to generate
- Examples: GTP…

Summarization



Encoder Transformer

Decoder Transformer

Encoder-Decoder Transformer

Summarization

Feature	BERT	GPT-1	T5
Model Type	Encoder-only	Decoder-only	Encoder-Decoder
Architecture	Bidirectional Transformer	Unidirectional Transformer (left-to-right)	Full Transformer (Encoder + Decoder)
Training Objective	Masked Language Modeling (MLM) + Next Sentence Prediction (NSP)	Causal Language Modeling (predict next token)	Span Corruption (denoise masked spans)
Input Format	Token sequence with random masks and segment embeddings (e.g., `[CLS] Sentence A [SEP] Sentence B [SEP]`)	Plain text sequence	Text-to-text format with task prefix (e.g., `"translate English to German: Hello"`)
Output Format	Contextualized embeddings / classification token (`[CLS]`)	Next token prediction	Text generation
Pretraining Data	BooksCorpus + Wikipedia	BooksCorpus	C4 (Colossal Clean Crawled Corpus)
Main Use Cases	Text classification, QA, NLI, embedding extraction	Language modeling, generation	Multi-task NLP: translation, QA, summarization, etc.
Release Year	2018	2018	2020

Summarization

- **Transfer Learning**: Source model --> knowledge --> Target model
- **Scaling law**: Powerful source model rely on big model trained with large-scale data!
- It is almost impossible to annotate large-scale data (e.g., billions of documents/images)
- We need self-supervised learning!

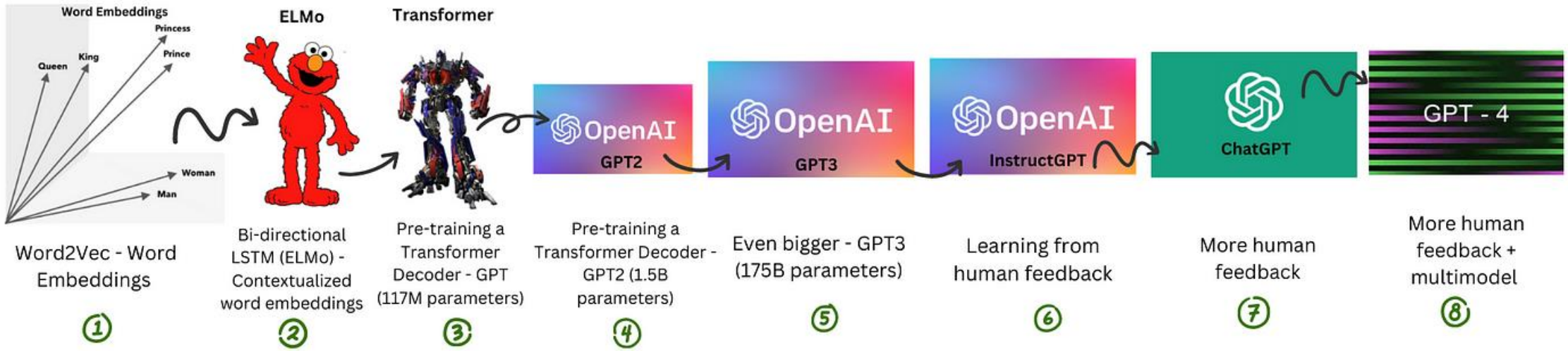
- Big models trained on unlabeled large-scale data in self-supervised way --> Pretrained Foundational Model

- On downstream tasks, we can use pretrained foundational models as backbone, and fine-tune it on small-scale task-specific data.

Pre-training and Fine-tuning paradigm!!

Pre-training and Fine-tuning Paradigm

- Later, transformer-based models such as ELMo, BERT, and GPT-1 extended this pretraining approach to trainable, transferable architectures for natural language tasks.
- It gradually became the core technological pathway for developing large language models.



https://medium.com/datareply/history-of-gpt-4-a71d4705b3d2

References

- Papers:
  - A Comprehensive Survey on Pretrained Foundation Models: A History from BERT to ChatGPT https://arxiv.org/abs/2302.09419
  - A Survey of Large Language Models https://arxiv.org/abs/2303.18223
  - Pre-trained Models for Representation Learning https://link.springer.com/chapter/10.1007/978-981-99-1600-9_5
- Online resources:
  - How to use pretrained models in HuggingFace: https://huggingface.co/learn/nlp-course/chapter4/2
  - https://blogs.nvidia.com/blog/what-are-foundation-models/
  - https://github.com/RUCAIBox/LLMSurvey?tab=readme-ov-file#llmsurvey
  - https://hackernoon.com/an-interview-with-ilya-sutskever-co-founder-of-openai
  - https://web.stanford.edu/class/cs224n/slides/cs224n-spr2024-lecture09-pretraining.pdf